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1 Chapter 10 — Population-Based Training and Evolutionary Game Theory: Experiment Report

Testbeds: four symmetric matrix games for the evolutionary-dynamics half (Prisoner’s Dilemma, Hawk-Dove, Rock-Paper-Scissors, Stag Hunt); a synthetic pure-skill ladder; the PSRO best-response meta-game on Leduc Hold’em; and a small AlphaStar-style **PBT league** of neural PPO agents playing Leduc Hold’em. The Leduc engine, the exact best-response oracle, and NashConv/exploitability come from the Chapter 07 stack (reused wholesale); PSRO and the meta-Nash solver come from Chapter 09. Neural agents are trained with PyTorch and then extracted to *tabular* policies so the Chapter 07 **exact** best response can grade them.

PhD connection: the population-level layer of the thesis. Three hooks: **main exploiters as automated opponent modelers** (Contribution #1, the population lift of Chapter 07’s Bayesian read); the **AlphaStar league as a population safe-exploitation mechanism whose guarantee is missing** (Contribution #2); and **EGTA / meta-Nash as the population evaluation methodology** (Contribution #3).

Scope of results: every number in this report is **measured from a real run** and read from the artifacts under `implementation/step10/implementation/results/{smoke,scale}_results.json` and `implementation/step10/exploration/figures/*.json`. Evolutionary results are bracketed by *analytic* references (ESS / Nash of each matrix game); league results are bracketed by Chapter 07’s *exact* exploitability. Three Phase-4 predictions were **contradicted** by the scale run; per WORKFLOW §0.1 they are kept as stated and reconciled with what actually happened (§8).

How to read this report. Both parts follow the same arc: **what we test -> how -> results -> conclusion**. **Part I (§1-§3)** uses evolutionary game theory as a *diagnostic lens*: replicator dynamics on solvable matrix games, then the transitive/cyclic “spinning-top” decomposition. **Part II (§4-§7)** builds and evaluates the **PBT league**: training dynamics, EGTA/meta-Nash, diversity + Elo, and a head-to-head against PSRO / self-play / CFR-Nash. §8 reconciles the three contradicted predictions; §9-§11 cover trust, limitations, and directions; §12 lists reproduction commands.

2 PART I — EVOLUTIONARY GAME THEORY AS A DIAGNOSTIC LENS

2.1 1. What this chapter is about

Chapters 2-9 reasoned about equilibria of a *fixed* game. Chapter 10 moves up a level to **populations of strategies that change over time** and asks two questions a single-agent view cannot: *which strategies grow?* (replicator dynamics) and *does this game even have a “best” strategy, or only a wheel of counters?* (the transitive/cyclic decomposition). These two tools are the diagnostic that Part II’s population-based training needs — because whether a self-training population converges or cycles is decided by the game’s structure, not the learning rate. All Part I evolutionary numbers are deterministic (fixed seeds, exact dynamics), so “did it converge?” is unambiguous.

2.2 2. Experiment 1 — replicator dynamics on four matrix games

What we test. Do the replicator dynamics reproduce each game’s *analytic* evolutionary outcome — convergence to a dominant strategy, to an interior ESS, to a basin-dependent pure ESS, or a non-convergent orbit? (Raw step validation, L530-533.)

How. Discrete-Euler replicator integration from a seeded interior start; “converged” means the last iterate stopped moving; “orbit radius” is the distance the trajectory holds from the uniform point. Analytic ESS/Nash is the reference. Data: `results/smoke_results.json` (replicator block), phase portraits in `exploration/figures/replicator_playground.json`.

Game	Analytic reference	Start x_0	Final x	Converged?	Orbit radius
Prisoner's Dilemma	Defect dominates; unique ESS (0, 1)	[0.666, 0.334]	[0.0, 1.0]	yes	0.707
Hawk-Dove	interior ESS $p(\text{Hawk}) =$ $V/C = 0.5$	[0.548, 0.452]	[0.5, 0.5]	yes	0.0
Rock- Paper- Scissors	Nash = uniform; no ESS (centre)	[0.347, 0.385, 0.268][0.299, 0.288, 0.413]		no	0.095
Stag Hunt ($x_0 =$ [0.8, 0.2])	two pure ESS (basin- dependent)	[0.8, 0.2]	[1.0, 0.0]	yes	0.707
Stag Hunt ($x_0 =$ [0.2, 0.8])	two pure ESS (basin- dependent)	[0.2, 0.8]	[0.0, 1.0]	yes	0.707

Results. All four match theory exactly. Prisoner's Dilemma collapses to all-Defect (the dominant strategy). Hawk-Dove converges to the interior mixed ESS at exactly 0.5 (orbit radius 0.0 — it *reaches* the rest point). Rock-Paper-Scissors **does not converge**: it orbits the uniform centre at a steady radius (0.095), because the interior fixed point is a centre, not an attractor. Stag Hunt lands on *different* pure ESS depending on the start — $[0.8, 0.2] \rightarrow \text{all-Stag}$, $[0.2, 0.8] \rightarrow \text{all-Hare}$ — the two basins made concrete.

Conclusion. Replicator dynamics are the continuous idealization of population selection, and their rest points are exactly Nash/ESS. The one that *refuses to converge* — RPS — is the whole reason Part II exists: a cyclic game has no stable population, so a self-training population in it will spin rather than settle.

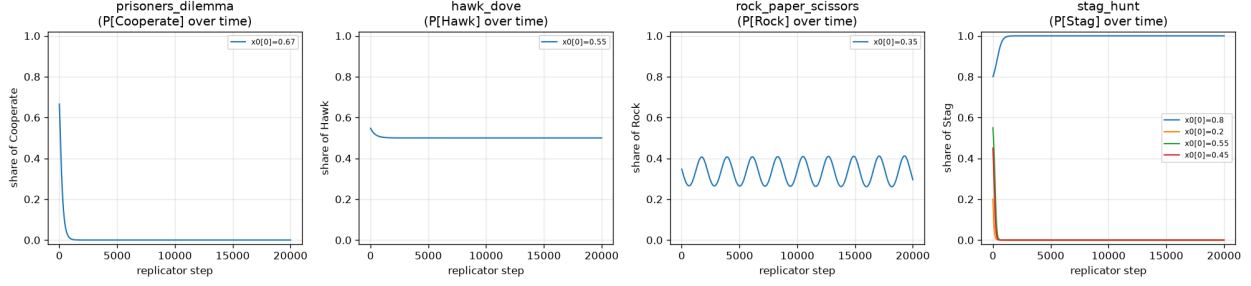


Figure 1: Replicator phase portraits on the four matrix games: Prisoner’s Dilemma collapses to all-Defect, Hawk-Dove converges to the interior 0.5 ESS, Rock-Paper-Scissors orbits the centre without converging, and Stag Hunt splits into two basins (all-Stag or all-Hare) depending on the start. The non-converging RPS orbit is the visceral case for the diversity machinery in Part II.

2.3 3. Experiment 2 — the spinning-top: transitive vs cyclic structure

What we test. Can we decompose a game’s payoff matrix into a **transitive** (skill-ladder) component and a **cyclic** (rock-paper-scissors) component, and does the decomposition correctly identify pure cases and diagnose real populations? (Raw step validation, L532, L534.)

How. The combinatorial-**Hodge** (ratings-difference) decomposition on the antisymmetrized payoff matrix, reporting a transitive ratio and a cyclic ratio (orthogonal, so they combine in quadrature to 1). We also report the raw-step’s SVD rank-1 transitive ratio for RPS to document its known failure. Data: `results/{smoke,scale}_results.json` (`spinning_top` block), `exploration/figures/game_landscape.json`.

Population	Transitive ratio (Hodge)	Cyclic ratio (Hodge)	Three-cycles	Note
Rock-Paper-Scissors	0.0	1.0	1	SVD rank-1 wrongly gives 0.707
Pure skill ladder	1.0	0.0	0	monotone strength
PSRO-Leduc meta-game (smoke, 8 rounds)	0.457	0.890	—	mostly cyclic
PSRO-Leduc meta-game (scale, 20 rounds)	0.411	0.912	27	mostly cyclic

Population	Transitive ratio (Hodge)	Cyclic ratio (Hodge)	Three-cycles	Note
League snapshot meta-game (smoke / scale)	0.981 / 0.937	—	—	mostly transitive

Results. The Hodge decomposition nails the pure cases: RPS is 0.0 transitive / 1.0 cyclic (a single three-cycle), the skill ladder is 1.0/0.0. The **SVD rank-1** method the raw step suggested instead reports RPS as ≈ 0.707 transitive — wrong, because a rank-1 skew-symmetric approximation cannot represent a pure cycle; the Hodge method is the one used for the diagnosis. The two *real* populations diverge sharply: the **PSRO best-response meta-game on Leduc is mostly cyclic** (≈ 0.41 - 0.46 transitive, 27 three-cycles), while the **league’s snapshot meta-game is mostly transitive** (≈ 0.94 - 0.98).

Conclusion. The transitive/cyclic ratio is a *pre-training diagnostic*. Two populations drawn from the same game (Leduc) have opposite structure depending on how they are built (best responses vs training snapshots — see §8.2). This is exactly the tool that predicts whether naive self-play/PBT will converge or cycle, and it is the diagnostic the thesis carries into later chapters.

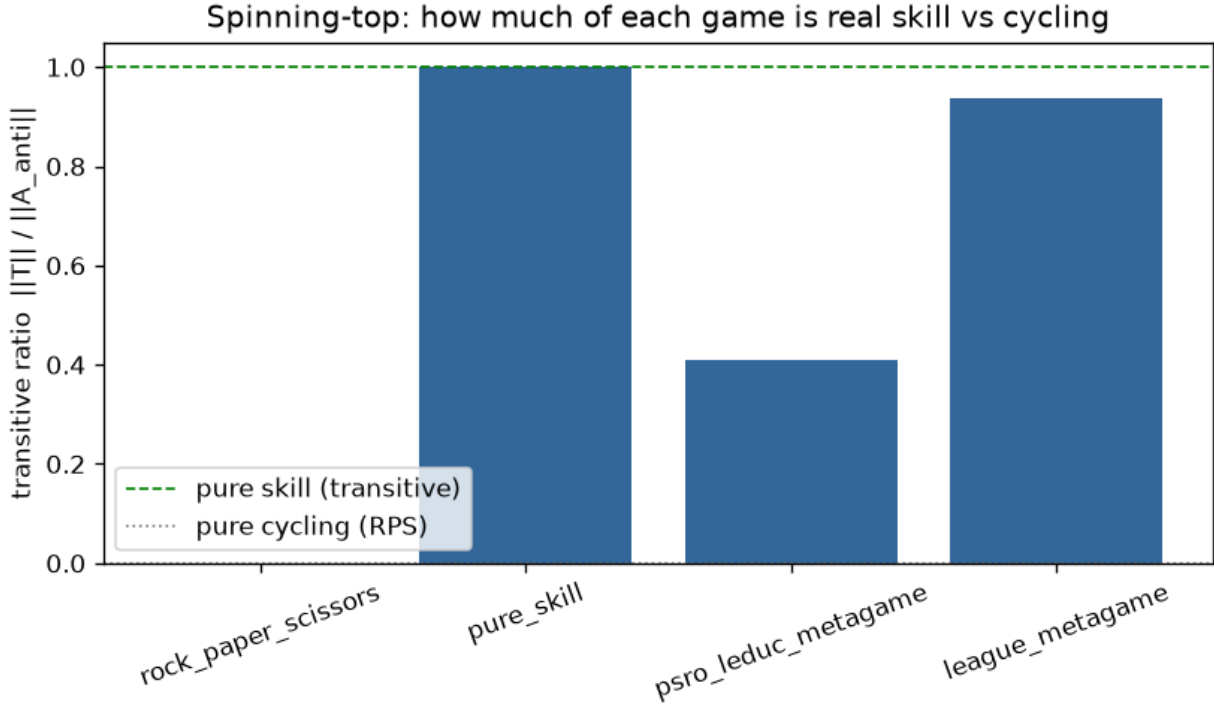


Figure 2: Transitive-ratio bar chart: Rock-Paper-Scissors (0.0, purely cyclic), a pure skill ladder (1.0, purely transitive), the PSRO-Leduc best-response meta-game (~ 0.41 - 0.46 , mostly cyclic with 27 three-cycles), and the league snapshot meta-game (~ 0.94 - 0.98 , mostly transitive). Which population you decompose decides whether Leduc looks like a wheel or a ladder.

3 PART II — THE PBT LEAGUE

3.1 4. What the league is and how it is graded

The second half builds a small **AlphaStar-style league** on Leduc Hold’em with three agent types — **main** agents (the product), **main exploiters** (hunt weaknesses in the current mains), and **league exploiters** (hunt weaknesses anywhere in the frozen history) — plus periodic **freezing** of snapshots into a museum and **PFSP** matchmaking (sample harder opponents more often). Agents are neural PPO networks on a Leduc info-state encoding; PBT copies the top agents (exploit) and perturbs their learning rate / entropy (explore). Crucially, each network is extracted to a **tabular** policy so Chapter 07’s **exact** best response measures its exploitability — the same NashConv yardstick used since Chapter 2. Two configs: **smoke** (7 live agents, 15 epochs) and **scale** (8 live agents, 120 epochs, 48 frozen snapshots).

3.2 5. Experiment 3 — league training dynamics

What we test. Does the league’s exploitability fall as training proceeds, and does the meta-Nash of the population track that improvement? (Raw step validation, L535, L537.)

How. Every epoch, record the minimum exploitability across the main agents and the exploitability of the current meta-Nash mixture; also record the league snapshot meta-game’s transitive ratio and final Elo. Data: `results/{smoke,scale}_results.json` (league block).

Metric	Smoke (15 epochs)	Scale (120 epochs)
min-main exploitability	4.67 → 3.04 (monotone, ends at min)	4.73 → min $\approx$ 1.21 (ep ~64) → 2.05 (ep 119)
meta-Nash exploitability	4.73 → 3.04	5.01 → min $\approx$ 1.32 (plateau 1.60) → 2.96 (ep 119)
league meta-game transitive ratio	0.981	0.937
final Elo (live agents)	1176-1211	1198-1210

Results. Smoke shows a clean monotone decline and ends at its minimum (3.04) — but 15 epochs is too short to show what happens next. Scale reveals it: exploitability falls steeply to a minimum near epoch 60-65 (**min-main** $\approx$ 1.21, meta-Nash bottoming $\approx$ 1.32 then holding a $\approx$ 1.60 plateau), and then **regresses** back up to $\approx$ 2.05 (min-main) and $\approx$ 2.96 (meta-Nash) by epoch 119. Elo spread is compressed (all live agents within $\approx$ 15 points), consistent with a behaviorally-similar population.

Conclusion. The league *does* drive strong early improvement — but improvement is **not monotone at scale**: under sustained exploiter pressure the live main agents chase their exploiters and lose ground on absolute exploitability late in training. The best agents are captured as *frozen snapshots* mid-run. This non-monotonicity is the honest headline of the scale run and is reconciled in §8.3.

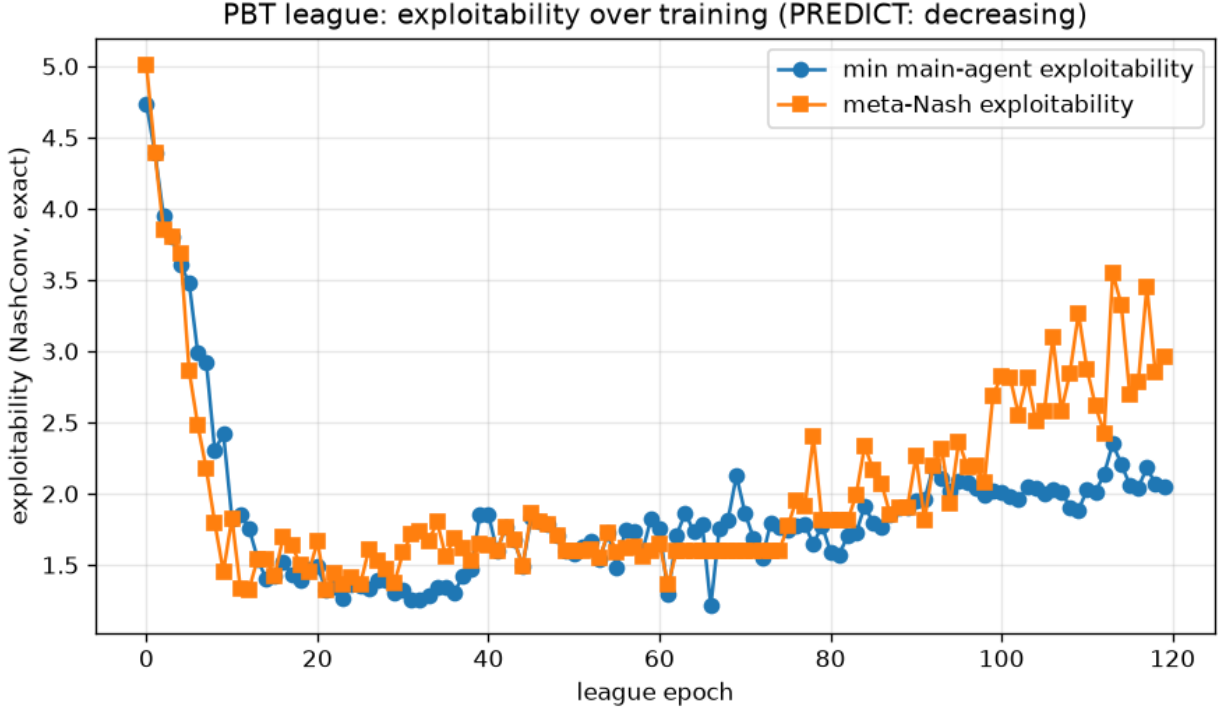


Figure 3: League exploitability over training (scale, 120 epochs): min-main and meta-Nash exploitability both fall steeply to a minimum near epoch 60 (min-main ~ 1.21 , meta-Nash ~ 1.32) and then regress upward (~ 2.05 / ~ 2.96) by epoch 119. A running league is not a monotonically improving one; the best agents are frozen snapshots from mid-run.

3.3 6. Experiment 4 — EGTA, meta-Nash, and diversity

What we test. (a) Is the meta-Nash of the final population less exploitable than any individual agent (the raw-step expectation, L536)? (b) How diverse is the population (effective size, behavioral clustering, exploit coverage — L538)?

How. Build the empirical symmetric payoff matrix over all agents (live + frozen), solve the meta-Nash mixture, collapse it to a single behavioral policy, and measure its exact exploitability against the best individual’s. Diversity: effective population (participation ratio of meta-Nash weights), single-linkage behavioral clustering, and exploit coverage. Data: `results/{smoke,scale}_results.json` (`league.egta`, `league.diversity`). Toy intuition: `exploration/figures/mini_pbt.json`.

(a) *Meta-Nash vs best individual:*

Config	meta-Nash exploitability	best individual	meta-Nash $\leq$ best?
Smoke	2.665	2.665	yes (all weight on the best agent)
Scale	3.418	1.305	no (weight spread 0.645+tail)

(b) *Diversity:*

Metric	Smoke	Scale
agents / active (participation)	16 / 1 (1.00)	56 / 3 (1.92)
behavioral clusters (thr 0.30)	1 (max pairwise 0.255)	1 (max pairwise 0.484)
exploit coverage	0.625	0.571

Results. (a) In smoke the meta-Nash puts all weight on the single best agent, so meta-Nash = best = 2.665 (the prediction holds trivially). At scale the meta-Nash spreads weight and its collapsed mixture is *more* exploitable (3.418) than the best single member (1.305) — the prediction **fails** (§8.1). (b) The population is only weakly diverse: participation ratio rises from 1 to 1.9, but behavioral clustering collapses everything into a **single cluster** at both scales (even though the scale population’s max pairwise distance 0.484 exceeds the 0.30 threshold — single-linkage merges the chain). The mini-PBT toy makes the mechanism vivid: on transitive Prisoner’s Dilemma diversity collapses to 0; on cyclic RPS it churns forever (0.07-0.29).

Conclusion. EGTA gives a population-level exploitability, but two lessons emerge: mixing behavioral policies via the meta-Nash does **not** guarantee a less-exploitable result (§8.1), and PBT diversity here is *weight-level*, not *behavior-level* — the agents are near-identical, so the “population” is thinner than its size suggests.

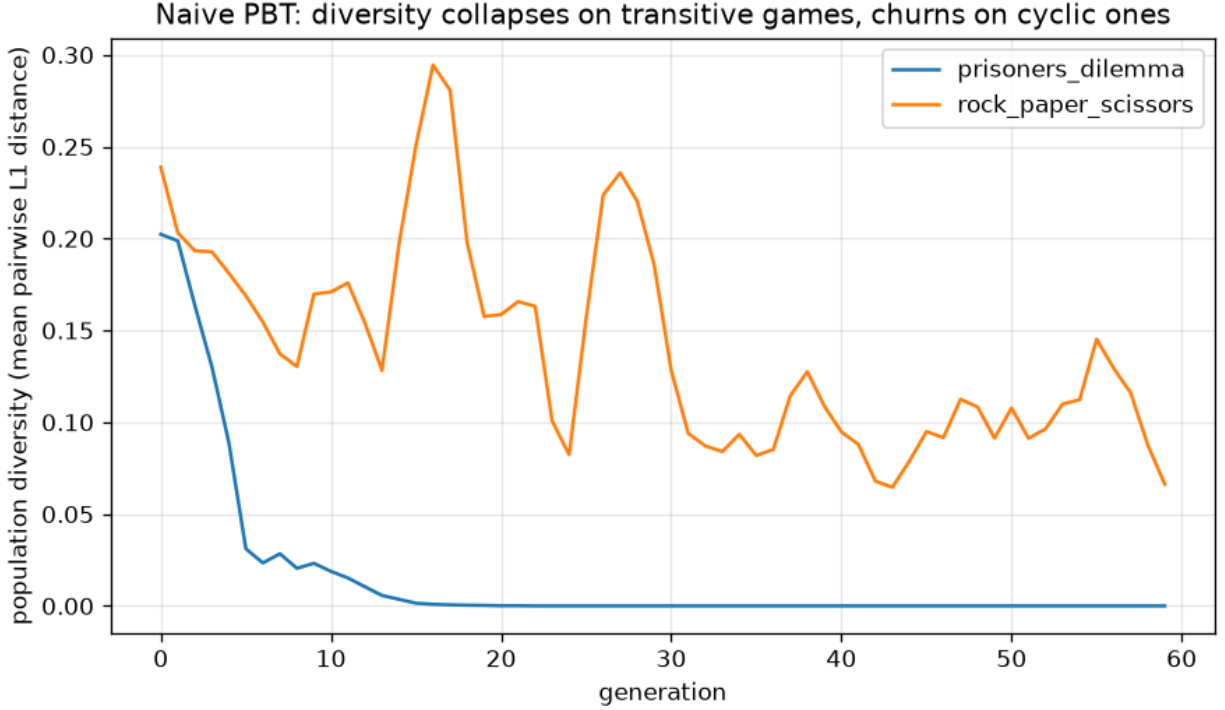


Figure 4: Mini-PBT diversity over generations: on the transitive Prisoner’s Dilemma the population’s strategy diversity collapses to zero (everyone converges to Defect), while on cyclic Rock-Paper-Scissors it churns indefinitely (0.07-0.29) as the population chases the wheel of counters. Game structure, not population size, sets whether diversity survives.

3.4 7. Experiment 5 — league vs PSRO vs self-play vs CFR-Nash

What we test. How does the league compare to the alternatives — exact PSRO, plain self-play, and CFR-Nash — on Leduc exploitability? (Raw step validation, L539.)

How. Run each method to its config’s budget and read the final exploitability. The league entry is its **best individual** (the strongest frozen snapshot); PSRO uses the exact BR oracle; CFR-Nash is the approximate-Nash floor. Data: `results/{smoke,scale}_results.json` (league, baselines).

Method	Smoke	Scale	Note
CFR-Nash (floor)	0.033 (2k iters)	0.0099 (20k iters)	approximate Nash, the target floor
League — best individual	2.665	1.305	strongest frozen snapshot
PSRO (exact BR)	3.037 (12 rounds)	2.163 (20 rounds)	Chapter 09’s slow-Leduc wall
Self-play	3.140	3.683	plain best-response-to-latest

Method	Smoke	Scale	Note
League — meta-Nash mixture	2.665	3.418	see §8.1

Results. At scale the league’s **best individual** (1.305) is the strongest of the learned methods — better than exact PSRO (2.163) and self-play (3.683) — though all remain far above the CFR-Nash floor (0.0099). The league’s **meta-Nash mixture** (3.418), by contrast, is the *weakest*, worse even than self-play (§8.1). Self-play actually *worsens* from smoke to scale, consistent with last-iterate non-convergence.

Conclusion. The league is a good producer of strong *individuals* — its best snapshot beats PSRO and self-play — but the *mixture* is not, and none of the learned methods approach the exact-CFR floor on Leduc. The right thing to ship from a league is a *selected member*, not the meta-Nash mixture (§8.1).

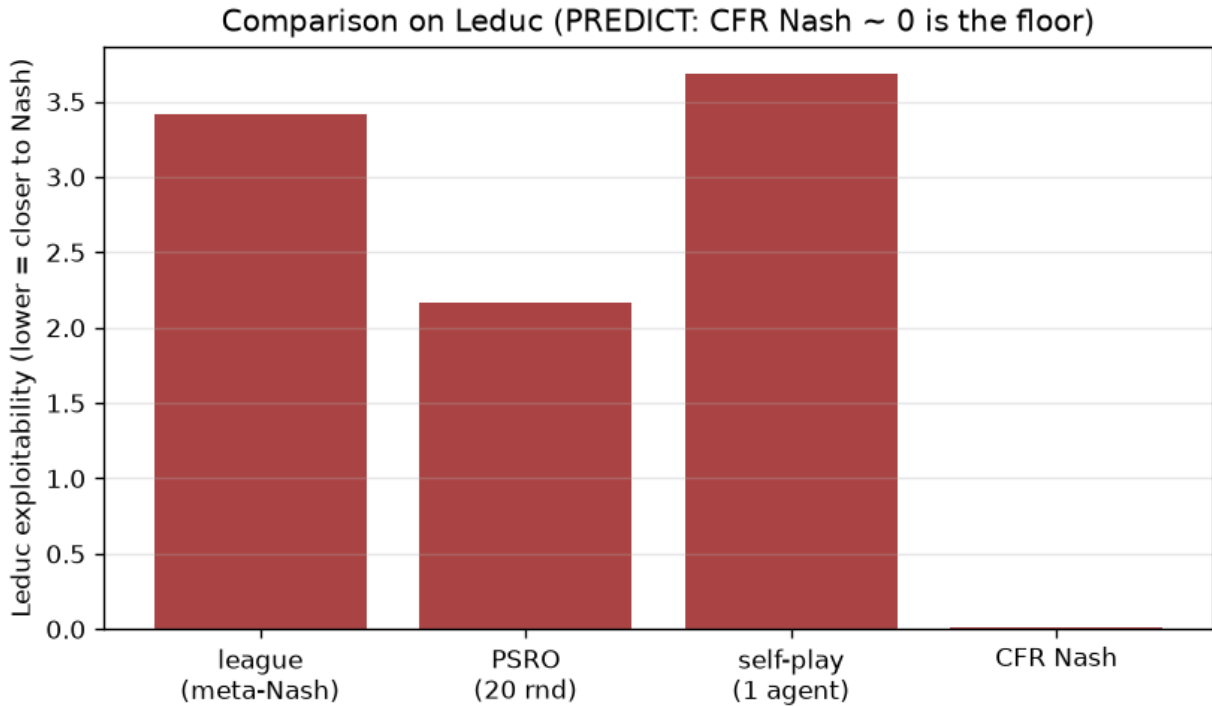


Figure 5: Final Leduc exploitability by method (scale): CFR-Nash floor ~0.01; the league’s best individual (1.31) beats PSRO (2.16) and self-play (3.68); the league’s meta-Nash mixture (3.42) is the weakest learned result. Ship a selected member, not the mixture.

3.5 8. Prediction <-> reality reconciliation

Per WORKFLOW §0.1, contradicted Phase-4 predictions are kept and reconciled, not silently edited. Three gaps.

1. **Meta-Nash of the league $\leq$ best individual (§6a).** *Predicted* (raw step Exit Checklist, L536): the meta-Nash is less exploitable than any individual agent. *Measured*: true in smoke ($2.665 = 2.665$, all weight on the best agent), **false at scale** ($3.418 > 1.305$). *Reconciliation*: not a mixing bug — the identical `mixture_behavioral_policy` code path gave meta = best in smoke. The meta-Nash minimizes **meta-game regret** (win against the population), which is a *different objective* from minimizing **full-game exploitability**; and a realization-weighted mixture of behavioral policies can be *more* exploitable than its components, because the mixture introduces info-set “tells” a best responder punishes. The lesson inverts and sharpens: population evaluation must measure the collapsed mixture’s exploitability directly and, in practice, ship a selected member rather than the mixture.
 2. **Leduc’s meta-game is transitive (§3).** *Predicted*: the intuition doc framed poker as a skill ladder. *Measured*: the PSRO **best-response** meta-game is mostly cyclic (≈ 0.41 - 0.46 transitive, 27 three-cycles), while the league **snapshot** meta-game is mostly transitive (≈ 0.94 - 0.98). *Reconciliation*: not a bug — a population of best responses cycles (A beats the mixture, B beats A, a later BR beats B; Balduzzi’s spinning-top), whereas a population of training-trajectory snapshots forms a ladder because later snapshots are mostly stronger. The structure is a property of the *population*, not just the game; both measurements are correct and instructive.
 3. **League exploitability decreases monotonically (§5).** *Predicted*: a monotone decline. *Measured*: smoke (15 epochs) declines cleanly and ends at its minimum (3.04); scale (120 epochs) falls to a minimum near epoch 60 (min-main ≈ 1.21 , meta-Nash ≈ 1.32 /plateau 1.60) and then **regresses** to ≈ 2.05 / ≈ 2.96 by epoch 119. *Reconciliation*: an honest, likely-real training instability, only visible once training is long enough — under sustained exploiter pressure the live main agents chase their exploiters and lose absolute-exploitability ground (churn / partial forgetting). The best agents are the frozen snapshots from mid-run, which is exactly why the best-individual metric (§7) is strong while the late live agents are not. Documented, not fixed; the natural remedy is best-snapshot retention / population regularization (§11). This echoes Chapter 09’s methodological rule: **scale reveals what smoke hides**.
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3.6 9. Trustworthiness and sample adequacy

- **Evolutionary results are bounded by analytic references.** Each replicator outcome is checked against the game’s analytic ESS/Nash; the spinning-top ratios are checked against the pure cases (RPS = 0.0, skill ladder = 1.0). These are deterministic and exactly reproducible.
 - **League exploitability is Chapter 07’s *exact* NashConv**, not a simulation: every neural policy is extracted to a tabular policy and scored by exact best response, so “how exploitable is this agent/mixture?” is ground-truthed rather than estimated.
 - **Neural training is seed- and library-sensitive.** The league is one PBT run per config; the *qualitative* findings (early improvement; best-individual < PSRO < self-play; late regression at scale; meta-Nash > best member at scale) are the trustworthy claims, not the third-decimal magnitudes.
 - **Diversity metrics are threshold-sensitive.** The single-cluster result depends on the 0.30 single-linkage threshold; it should be read as “behaviorally similar,” not “provably one strategy.”
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3.7 10. Limitations (ranked by how much they affect the conclusions)

1. **Late-training regression at scale (§5, §8.3)** — the league does not monotonically improve; until best-snapshot retention/regularization is added and re-run, the “league improves” claim holds only for the *first half* of training and for the *frozen best*, not the live agents.
 2. **Meta-Nash mixture more exploitable than the best member (§6a, §8.1)** — means population evaluation must not report the mixture as the league’s strength; the selected member is the right artifact.
 3. **Thin behavioral diversity (§6b)** — participation ratio 1.9 and a single behavioral cluster mean the “population” is near-degenerate; the diversity benefits AlphaStar reports at ~600 agents do not appear at this scale.
 4. **Single-run neural results (§9)** — direction-robust, magnitude-uncertain.
 5. **Toy scale throughout** — Leduc and small matrix games; nothing here should be extrapolated to deep-RL leagues at AlphaStar scale.
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3.8 11. Conclusions and research directions

Conclusions. Part I delivers the diagnostic: replicator dynamics reproduce every analytic ESS/Nash (including RPS’s non-converging orbit), and the Hodge spinning-top decomposition cleanly separates skill from cycles — revealing that the *same* game (Leduc) is cyclic as a best-response population but transitive as a snapshot population. Part II builds the league and

grades it with exact exploitability: it produces strong *individuals* (best snapshot 1.305, beating PSRO 2.163 and self-play 3.683), but three predictions broke honestly — the meta-Nash mixture is *more* exploitable than its best member at scale, the Leduc best-response meta-game is *cyclic* not transitive, and league exploitability *regresses* late in long training. The population machinery works, but carries no monotonicity or safety guarantee — which is precisely the thesis gap.

Research directions (each tied to a measured effect):

- *Best-snapshot retention / population regularization* — directly targets the late regression (§8.3); re-run to see whether the guarantee gap is a training or a design issue.
- *Report the selected member, or a diversity-regularized meta-solver* — the meta-Nash-mixing failure (§8.1) motivates either shipping a best-response-robust member or changing the meta-solver objective.
- *Carry the transitive/cyclic diagnostic to Chapter 11’s FFA games* — test the “large cyclic component” prediction directly, where naive PBT is expected to cycle.
- *Formalize population safety without a minimax anchor (Contribution #2)* — the exploiter mechanism is heuristic; §8.1/§8.3 show it does not guarantee a non-exploitable population.

3.9 12. Reproduction

From `implementation/step10/implementation/` with the project `.venv` active:

```
# validation harness (PASS/FAIL/SKIP against the raw-step targets)
python validate.py
```

```
# full comparison runs -> results/{smoke,scale}_results.json
python tournament.py --config smoke
python tournament.py --config scale
```

```
# plots from a results JSON -> plots/*.png (needs matplotlib)
python plotting.py --config scale
```

From `implementation/step10/exploration/` (Part I figures + JSON):

```
python replicator_playground.py      # replicator phase portraits + JSON
python game_landscape.py              # transitive/cyclic ratios (RPS / skill / PSRO-Leduc)
python mini_pbt.py                   # diversity collapse vs churn
python psro_population_peek.py        # PSRO population dynamics on Leduc
```

Seeds are fixed in `config.py` (implementation) and each exploration script’s config. The evolutionary and spinning-top results are exactly reproducible; the neural league results are direction-stable

across seeds. All numbers in this report were read from `results/{smoke,scale}_results.json` and `exploration/figures/*.json`.